

INTERNSHIP REPORT

NEELUM JEHLUM HYDROPOWER COMPANY
SITE C3 CHATAR KALAS

SUBMITTED TO

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CONTENTS

- INTRODUCTION
 - ✓ BACKGROUND
- POWER HOUSE
- TURBINE SYSTEM
 - ✓ PENSTOCK
 - ✓ MAIN INLET VALVE
 - ✓ BYPASS VALVE
 - ✓ SPIRAL CASING
 - ✓ WICKET GATE
 - ✓ RUNNER
 - ✓ BEARINGS
 - ✓ HEAD COVER
 - ✓ SHAFT
 - ✓ ROTOR
 - ✓ DRAFT TUBE
- AUXILIARIES
 - ✓ COOLING WATER SYSTEM
 - ✓ MP COMPRESSORS
 - ✓ BRAKE COMPRESSORS
 - ✓ INDUSTRIAL COMPRESSORS
 - ✓ DRAINAGE PUMP
 - ✓ UNIT DEWATERING PUMPS + DUTY PUMPS
 - ✓ TURBINE OIL SYSTEM
 - ✓ IPB COMPRESSORS
 - ✓ HRT DEWARTING SYSTEM
 - ✓ TAILRACE DEWATERING SYSTEM
 - ✓ FLAP GATE
 - ✓ GOVERNOR SYSTEM OF POWER HOUSE
 - ✓ CO₂ FIRE FIGHTING SYSTEM
 - ✓ CRANES
- CONCLUSION

INTRODUCTION

BACKGROUND

Project is running through Neelum Jhelum Hydro-Power Company (NJHPC) WAPDA, headed by a board of Directors. The Chairman of BOD is Chairman Wapda. Neelum Jehlum Hydroelectric Project is located in the vicinity Muzaffarabad (AJ&K). It envisages the diversion of Neelum river water through a tunnel out-falling into Jhelum river. The intake Neelum Jehlum is at Nauseri 41 km East of Muzaffarabad. The Power House is at Chatter Kalas, 22 Km South of Muzaffarabad. After passing through the turbines the water is released into Jhelum River about 4 Km South of Chatter Kalas. Neelum Jhelum Hydroelectric Project has installed capacity of 969 MW. The Project is produce 5.15 Billion units of electricity annually. A Composite Dam (Gravity+ Rock fill) 160 m long and 60 m high had been constructed on Neelum River at Nauseri. It is a Gated Diversion Dam. The total length of head race tunnel is about 48 km. A 19.54 km stretch of the tunnel from the Nauseri site is constructed as a twin tunnel system each with cross sectional area ranging from 52-58m². The remaining headrace tunnel down to the surge chamber is a single tunnel having cross sectional area 100m² approximately. The Surge Chamber consist of 341m high riser shaft and 820m long surge tunnel, four steel lined Penstock tunnels 118 m long and having 3.8 m internal diameter will also be constructed. The underground power Station will have four units with a total capacity of 969 MW. The Power Station is connected with Gakhar Grid station through 525 KV double circuit transmission line.

The Neelum Jhelum Hydroelectric Project is split into the following three main geographic areas



Powerhouse

Water passes the Headrace Tunnel from Dam site (C1) to Underground Powerhouse (C3) for generation of power. Here is the general overview that how turbine system and main Auxiliaries work together and how they are connected to Transformer Hall and Switch Yard. Powerhouse is mainly comprised of following parts:

Turbine System

Here at powerhouse the turbine used is Francis Turbine. It is an inward flow Reaction Turbine that combines radial and axial flow concepts.

1) Penstock 2) Main Inlet Valve 3) Bypass Valve 4) Spiral Casing 5) Wicket Gates 6) Runner 7) Bearings 8) Head Cover 9) Shaft 10) Rotor 11) Draft Tube

PENSTOCK

A penstock is an enclosed pipe that delivers water from headrace tunnel to Main Inlet Valve. So the first thing in which the water enters in Powerhouse is Penstock.

Main Inlet Valve

This valve comes after the water passes from penstock. It is basically to control the flow of water. That is either the water is allowed to pass to hit the runner or not. It is of Ball-Valve type. That is spherical but with holes at two sides. It is closed to stop the flow of water when unit is closed or when maintenance is needed. We open or close it manually using counter weights which uses hydraulics system.

Bypass Valve

It is fixed adjacent to Main Inlet Valve across its inlet and outlet. Its purpose is basically, when the Main Inlet Valve is closed there is a pressure of water in its inlet side but there is no pressure in outlet side, so we have to equalize pressure on both sides first to avoid components damages after the valve is opened and hence for this purpose bypass valve is used.

Spiral Casing

Here in Powerhouse Francis Turbine is used. So Spiral casing is an important component for this type of turbine for even distribution of kinetic and potential

energy of water so as to achieve the required reaction for better performance of the runner. Its purpose is basically to enable water to hit runner radially so that it can create pressure difference across the two faces of runner blade as a reaction of which runner rotates.

Wicket Gate

It is also called guide vane. Its purpose is basically to control the degree of flow of water. That is what percentage of total water passed from Main Inlet Valve should be allowed to hit the runner. The control of wicket gate opening and closing allows the output energy generated by the turbines to be controlled to match the desired energy output. It is governed by Oil Governor System. A series of small openings of the wicket gates surround the turbine. When they are opened wider, they allow more water to pass and vice versa. So their angle defines the amount of water to be passed through. Here the maximum allowed capacity of the water to be passed through is 85% of the total water.

Runner

Here at Powerhouse Francis turbine has been used. It basically rotates using the flow of water and hence making shaft to rotate, which rotates rotor to generate power output. So for this purpose Runner blades are the heart of any turbine. These are the centres where water strikes and the tangential force of impact causes the shaft of the turbine to rotate. When water passes from spiral casing it enters the runner radially and exits through it axially making applying the torque at the runner blade hence forcing it to rotate.

Bearings

The generator has upper and lower guide bearings and one (1) thrust bearing. The upper and lower guide bearings are located at in the oil tank of the centre body of upper and lower brackets individually. The thrust bearing locates above rotor, in the oil tank above upper bracket.

Thrust bearing

The thrust bearing above the upper bracket, adopt the structure of segments supported by flexible oil tanks. It has 10 pads. Pads have high-pressure oil cavity. The 81 high-pressure oil lubrication system is used for supply high-pressure oil to the thrust bearing pads at generator when the unit starts and stops. The total load of the thrust bearing is about 730 ton. The thrust bearing bears the weight

of all rotating part of the turbine generator unit and the axial load created by water thrust.

Guide bearing

The generator is equipped with upper guide bearing and lower guide bearing. The upper and lower guide bearings are arranged inside the centre body of the upper generator bracket and the lower generator bracket respectively. The guide bearing is an oil immersed structure composed of Babbitt alloy bushes. The bush back is provided with spherical support capable of transmitting radial force to the upper generator bracket. The shims with different thicknesses are provided between the spherical support and the back of the bush for adjusting the assembling clearance between the guide bearing bush and the guide bearing collar.

Head Cover

It is above the runner. When water exits runner it has some kinetic energy and pressure left and hence to avoid it to go to the shaft head cover is used above the runner.

Shaft

It is basically a metallic rod that connects runner to the rotor. When runner rotates it produces torque. Shaft transmits that torque to the rotor and hence making it to rotate. It is connected to Runner and Rotor through bearings.

Rotor

Rotor is the last mechanical component of turbine. As stated earlier, shaft transfers torque to rotor so it rotates. Then its rotation is used to convert its mechanical energy to electrical energy using Generator system.

Draft Tube

It is basically a conduit that connects the runner exit to the tail race where the water is discharged from the turbine. Its main function is to reduce the velocity of discharged water to minimize the loss of kinetic energy at the outlet. This permits the turbine to be set above the tail water without appreciable drop of available head.

AUXILIARIES

Water Cooling System

As we know due to friction among the rotating parts, heat is generated such as in bearings that are holding the main components of the turbine together. So in order to accumulate the heat generation which can cause damage to bearing and other parts as well, Water Cooling System is used.

First extracting the water from the outlet of turbine in the draft tube region. The water is then supplied in the cooling system via pumps where it is filtered before and after the pumping mechanism to avoid clogging the system. The water is then transferred to the cooling system via insulated pipes.

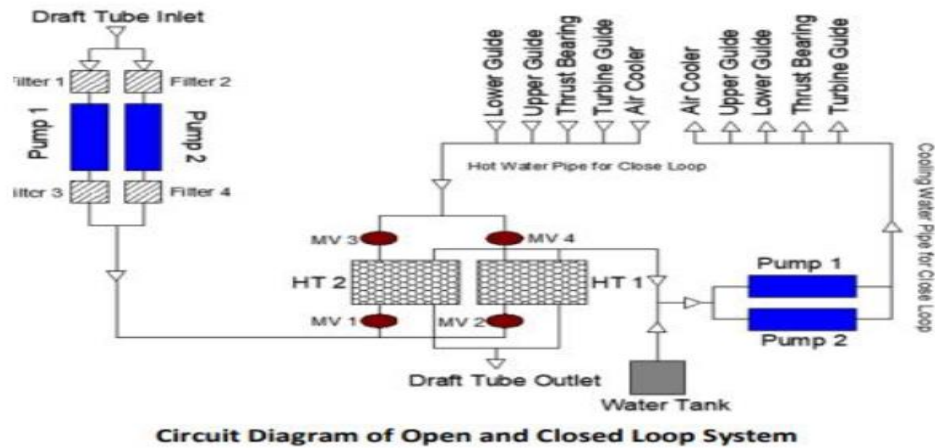
Here basically there are two loops involved in the cooling system i.e. open loop and close loop. Open loop involves discharge of water after heat transfer whereas the close loop reuses the same water.

In addition to this there is MT (Main Transformer) Cooling System as well. Its working mechanism is same, with a difference that it is used for cooling of coils or other components of Transformer.

Here water is circulated in the pipes using Centrifugal Pumps. They are used to transport fluids by the conversion of rotational kinetic energy to the hydrodynamic energy of the fluid flow. The rotational energy typically comes from an engine or electric motor. Cooling is done using Heat Exchangers. Heat Exchangers use cool water to absorb heat from the hot oil or any fluid by passing it through the oil using either

Tube-Shell Heat Exchanger or a Plate Heat Exchanger. Here Shell and Plate type Heat Exchangers are being used

Water Storage Tank is also used along with pumps to store water to be circulated. There is also a pressure valve on a storage tank, whose function is to maintain water level in the close loop. Hot water from bearings and generator enters the close loop and inserted into the heat exchangers for cooling. The exchange of heat occur between the hot water of close loop and the cold water from open loop hence close loops water temperature decreases and open loops water temperature increases. MV are motorized valves present at inlet which can be remotely/automatically controlled. The open loop water is then discharged back to the draft tube and close loop water is pumped back into the system to maintain temperatures in bearings and generator.



MP Compressors

It is a three-stage piston type compressor. Power of compressor is 15 KW. Speed and pressure produced by compressor is 500 rpm and 7.78 MPa respectively. It is used for main inlet valve and governor system of the power house.

Brake Compressors

High pressure compressed air system for governing control system of generating units is provided common for all the units. Compressed air systems comprises of two nos. suitable capacity high pressure compressors one working as main and the other for stand by use, one common high pressure air receiver for supplying high pressure compressed to governing pressure vessels and one no. low pressure air receiver for supplying low pressure air for generator breaks and for use of pneumatic tools during maintenance operation.

Although it also use for maintaining pressure in the regulator tank. Power is 15 KW, rated pressure is 0.85 MPa and capacity is 2.36 m³/min.

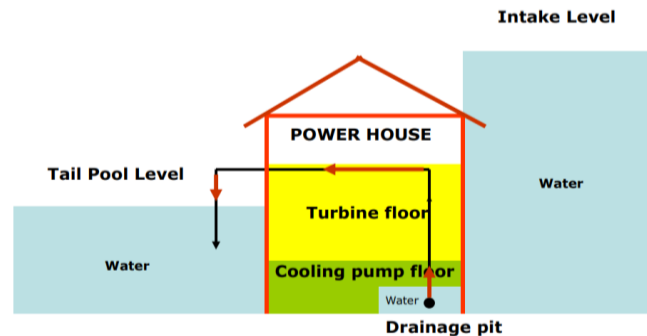
Industrial Compressors

Capacity of these compressors is 9.1 m³/min. Pressure and power of these are 0.85 MPa and 55 KW respectively. These are screw type compressors and It is used for maintenance services of the powerhouse.

Drainage Pump

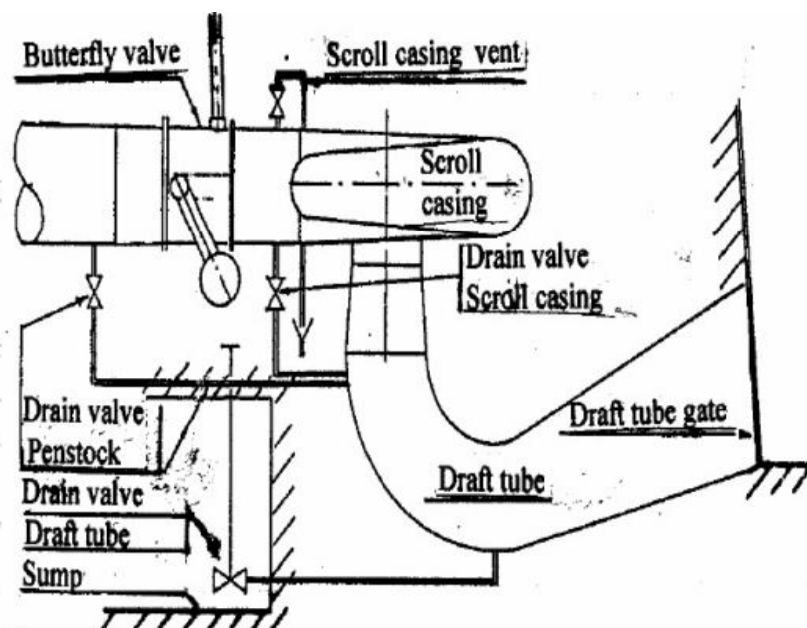
The drainage pump is used for draining pits and deep roadway underpasses, as well as for dewatering basements and courtyards subject to a flooding risk or groundwater. It is usually started automatically via float switches. The water to be handled may contain dirt and solids whose permissible particle size depends

on the pump size and impeller. Though, there is seepage in the power house and water goes into the drain where through drainage pumps it goes into tailrace tunnel.



Unit Dewatering System

The dewatering system is generally meant to dewater the penstock, main inlet valve, spiral case and runner during their maintenance and inspection works.



Turbine Oil System

For large & medium turbines this system consists of a pumping set comprising two or more positive displacement pumps drawing oil from sump and delivering it passed unloading and safety valves, through a non return valve, and into an oil/air pressure oil vessel or pressure receiver whence it is fed to the Governor actuator & main servo motor regulating valve. Exhaust oil from servo motor is return to the sump.

Storage Equipment:

Pressure Tank used to store pressure and reduce hydraulic shock.

Air/Oil type:

Consist of a fluid at the bottom and a compressed air at the top without any medium for separation.

IBP Compressors

Isolated-phase bus (IPB), also known as is a method of construction for circuits carrying very large currents, typically between in a generator and its step up transformer in a steam or in a large hydroelectric power plant. IBP compressors extract moistures and humidity from the atmosphere for preventing the high current lines and give clean environment. Compressor power is 15 KW and its speed is 500 rpm. Discharge pressure is 8.0 MPa.

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Governor System of Powerhouse

Hydraulic turbine governor is used for controlling the guide vanes by detecting turbine speed and its guide vane opening in order to keep turbine speed stable or to regulate its output. The Performance of the governor dominates the controllability of the power plant and quality of an electrical power produced.

Flap Gate

Hydro Gate® flap gates are made of cast iron or ductile iron, depending on the type of service. A small differential pressure on the back of the gate causes it to open automatically to allow discharge through levees, sewer lines or drainage conduits. When water on the face side of the gate rises above water on the back side, the gate closes automatically to prevent backflow. Flap gates are equipped with flat-back seats for attaching to wall thimbles, new concrete headwalls, existing walls or pipe flanges. The seat or frame of the flap gate is attached to a

wall or pipe flange and forms the opening through which water passes. Since the gate opens or closes automatically, a mechanical lifting device is not necessary.

Cranes

The main traveling crane in a Power station plays a very large part in the initial installation of Plant and thereafter periodic overhaul of the main machines. The principle lifts may be as much as 450 Ton but it is important that the crane should also have convenient facilities for dealing quickly with small loads, and for this purpose it is usual to provide a high speed auxiliary hook capable of handling about five to twenty five tons. The longitude travel and transfer movements are usually motor operated.

CO₂ Fire-Fighting System

A fire needs oxygen, fuel, and heat to continue to burn. By eliminating one of these three elements, a fire can be suppressed or extinguished. While some suppression agents reduce the heat of the fire, a CO₂ fire suppression system eliminates the oxygen to suppress the fire. When the suppression system detects smoke or fire, it then releases the CO₂ agent into the space it is protecting. The CO₂ level in the space quickly increases as the oxygen level quickly drops causing the fire to be suppressed or extinguished.

CONCLUSION

This Internship was really helpful in making me understand the working of engineering bodies at field , a hand on experience of how things works at the site. I had the opportunity to see in person the mechanical engineering techniques being applied in field which I was taught about in my lectures. My perception of mechanical engineering techniques from what I learned from the books was evolved into a better, professional knowledge of field. This internship which was basically to learn about the Neelum Jhelum Hydroelectric Plant and the different Engineering Techniques being used at the project taught us.